第十三章：《古典概型与古典概型概率计算》知识点教材

Textbook on Knowledge Points of Classical Probability Model and Its Probability Calculation

一、事件的关系与运算（古典概型计算基础）

I. Relationships and Operations of Events (Foundation of Classical Probability Calculation)

1. **包含关系**：若事件R的所有样本点都属于事件R_1（即$R \subseteq R_1$），则称事件R_1包含事件R。

Inclusion Relationship: If all sample points of event R belong to event R_1 (i.e., $R \subseteq R_1$), then event R_1 is said to contain event R.

2. **互斥事件**：若事件R与事件G没有公共样本点（即$R \cap G = \emptyset$），则称两事件互斥，意味着它们不能同时发生。

Mutually Exclusive Events: If events R and G have no common sample points (i.e., $R \cap G = \emptyset$), the two events are said to be mutually exclusive, meaning they cannot occur simultaneously.

3. **对立事件**：若事件M与事件N满足$M \cup N = \Omega$（样本空间）且$M \cap N = \emptyset$，则称两事件互为对立事件，意味着它们必有一个发生且只有一个发生。

Complementary Events: If events$M$ and N satisfy $M \cup N = \Omega$ (sample space) and $M \cap N = \emptyset$, the two events are said to be complementary to each other, meaning exactly one of them must occur.

4. **并事件**：若事件M由事件R和事件G的所有样本点组成（即$R \cup G = M$），则称M是R与G的并事件，表示"至少一个发生"。

Union Event: If event M consists of all sample points of events R and G (i.e., $R \cup G = M$), then M is called the union event of R and G, indicating "at least one occurs".

5. **交事件**：若事件R由事件R_1和事件R_2的公共样本点组成（即$R_1 \cap R_2 = R$），则称R是R_1与R_2的交事件，表示"同时发生"。

Intersection Event: If event R consists of the common sample points of events R_1 and R_2 (i.e., $R_1 \cap R_2 = R$), then R is called the intersection event of R_1 and R_2, indicating "both occur simultaneously".

二、古典概型的定义与特征

II. Definition and Characteristics of Classical Probability Model

1. **定义**：具有特定特征的随机试验称为古典概型试验，其数学模型称为古典概率模型（简称古典概型）。

Definition: A random experiment with specific characteristics is called a classical probability experiment, and its mathematical model is called the classical probability model (referred to as classical probability model for short).

2. **核心特征**：**Core Characteristics**:

- 有限性：样本空间Ω包含的样本点个数是有限的；

  Finiteness: The number of sample points contained in the sample space Ω is finite;
- 等可能性：每个样本点发生的概率大小完全相等。

  Equiprobability: Each sample point has exactly the same probability of occurring.

三、古典概型的概率公式

III. Probability Formula of Classical Probability Model

1. **公式推导**：设古典概型试验的样本空间Ω包含n个样本点（记为$n(\Omega)=n$），事件A包含其中k个样本点（记为$n(A)=k$）。由于每个样本点等可能发生，事件A的概率为：

Formula Derivation: Let the sample space Ω of a classical probability experiment contain n sample points (denoted as $n(\Omega)=n$), and event A contain k of these sample points (denoted as $n(A)=k$). Since each sample point occurs with equal probability, the probability of event A is:

$$P(A)=\frac{k}{n}=\frac{n(A)}{n(\Omega)}$$

2. **历史说明**：该公式由法国数学家拉普拉斯（P.-S. Laplace）在 1812 年作为概率的一般定义提出，称为概率的古典定义。

Historical Note: This formula was proposed by the French mathematician P.-S. Laplace as the general definition of probability in 1812, known as the classical definition of probability.

四、古典概型概率计算的一般步骤

IV. General Steps for Calculating Classical Probability

1. **明确试验与样本点**：用字母、数字或数组等符号表示试验的所有可能结果（样本点），确保不重不漏（可借助图表辅助列举）；

Clarify the Experiment and Sample Points: Use symbols such as letters, numbers, or arrays to represent all possible outcomes (sample points) of the experiment, ensuring no repetition or omission (enumeration can be assisted by charts);

2. **验证古典概型特征**：判断样本点是否满足“有限性”和“等可能性”；

Verify the Characteristics of Classical Probability Model: Determine whether the sample points satisfy "finiteness" and "equiprobability";

3. **计算样本点总数**：统计样本空间Ω包含的样本点个数$n(\Omega)$；

Calculate the Total Number of Sample Points: Count the number of sample points $n(\Omega)$ contained in the sample space Ω;

4. **计算事件含有的样本点个数**：统计目标事件A包含的样本点个数$n(A)$；

Calculate the Number of Sample Points Contained in the Event: Count the number of sample points $n(A)$ contained in the target event$A$;

5. **代入公式计算**：利用$P(A)=\frac{n(A)}{n(\Omega)}$求出概率。

Substitute into the Formula for Calculation: Calculate the probability using$P(A)=\frac{n(A)}{n(\Omega)}$.

五、典型例题（含解析）

V. Typical Examples (with Explanations)

例 1 单项选择题概率

Example 7 Probability of Multiple-Choice Questions

题目：标准化考试中，单选题从A,B,C,D四个选项中选一个正确答案。若考生随机选择答案，答对的概率是多少？

Question: In a standardized exam, a multiple-choice question has four options A, B, C, and D, with one correct answer. What is the probability that a candidate answers correctly by randomly choosing an option?

解析：

Explanation:

- 样本空间$\Omega=A,B,C,D$，共$n(\Omega)=4$个样本点，且每个选项被选的可能性相等（满足古典概型特征）；

The sample space $\Omega=\{A,B,C,D\}$ has a total of $n(\Omega)=4$ sample points, and each option is equally likely to be chosen (satisfying the characteristics of the classical probability model);

- 设$M=$“选中正确答案”，由于正确答案唯一，$n(M)=1$；

Let $M=$ "selecting the correct answer". Since there is only one correct answer, $n(M)=1$;

- 概率计算：$P(M)=\frac{n(M)}{n(\Omega)}=\frac{1}{4}$。

Probability calculation:$P(M)=\frac{n(M)}{n(\Omega)}=\frac{1}{4}$.

例 2 抛掷两枚骰子的概率

Example 8 Probability of Tossing Two Dice

题目：抛掷两枚质地均匀的骰子（标记为 I 号和 II 号），完成下列问题：

Question: Toss two fair dice (marked as Die I and Die II) and complete the following questions:

(1) 写出样本空间并判断是否为古典概型；

(1) Write the sample space and determine if it is a classical probability model;

(2) 求事件$A=$“两个点数之和是 5”、$B=$“两个点数相等”、$C=$“I 号骰子点数大于 II 号骰子点数”的概率。

(2) Find the probabilities of event $A=$ "the sum of the two points is 5", $B=$ "the two points are equal", and $C=$ "the point of Die I is greater than that of Die II".

解析：

Explanation:

(1) 用(m,n)表示 I 号骰子点数为m、II 号骰子点数为n，样本空间$\Omega = (m,n) \mid m,n \in$ $1,2,3,4,5,6$，共36个样本点，各样本点等可能发生，为古典概型；

(1) Let (m,n) represent that the point of Die I is m and the point of Die II is n. The sample space $\Omega = \{(m,n) \mid m,n \in \{1,2,3,4,5,6\}\}$ has a total of 36 sample points, and each sample point occurs with equal probability, so it is a classical probability model;

(2) ① $A = (1,4),(2,3),(3,2),(4,1)$，$n(A) = 4$，$P(A) = \frac{4}{36} = \frac{1}{9}$；

② $B = (1,1),(2,2),(3,3),(4,4),(5,5),(6,6)$，$n(B) = 6$，$P(B) = \frac{6}{36} = \frac{1}{6}$；

③ $C = \{(2,1),(3,1),(3,2),\dots,(6,5)\}$，$n(C) = 15$，$P(C) = \frac{15}{36} = \frac{5}{12}$。

例 3 不放回摸球概率

Example 9 Probability of Drawing Balls Without Replacement

题目：袋子中有 5 个球（2 红 3 黄），不放回依次摸出 2 个球，求：

Question: There are 5 balls in a bag (2 red and 3 yellow). Two balls are drawn in sequence without replacement. Find:

(1) A =“第一次摸到红球”；(2) B =“第二次摸到红球”；(3) AB =“两次都摸到红球”的概率。

(1) Probability of A = "drawing a red ball first"; (2) Probability of B = "drawing a red ball second"; (3) Probability of AB = "drawing red balls twice in a row".

解析：

Explanation:

- 红球编号 1,2，黄球编号 3,4,5，样本空间共$5 \times 4 = 20$个样本点；Number the red balls 1 and 2, and the yellow balls 3, 4, and 5. The sample space has a total of $5 \times 4 = 20$ sample points;
- (1) A包含 8 个样本点，$P(A) = \frac{8}{20} = \frac{2}{5}$；(1) Event A contains 8 sample points, so $P(A) = \frac{8}{20} = \frac{2}{5}$;
- (2) B包含 8 个样本点，$P(B) = \frac{8}{20} = \frac{2}{5}$；(2) Event B contains 8 sample points, so $P(B) = \frac{8}{20} = \frac{2}{5}$;
- (3) AB包含 2 个样本点，$P(AB) = \frac{2}{20} = \frac{1}{10}$。(3) Event AB contains 2 sample points, so $P(AB) = \frac{2}{20} = \frac{1}{10}$.

例 4 不同抽样方式下的概率

Example 10 Probability Under Different Sampling Methods

题目：从 2 名男生（B_1,B_2）和 2 名女生（G_1,G_2）中抽 2 人，分别计算有放回、不放回、分层抽样中“抽到两名男生”的概率。

Question: Draw 2 people from 2 boys (B_1,B_2) and 2 girls (G_1,G_2). Calculate the probability of "drawing two boys" under with-replacement sampling, without-replacement sampling, and stratified sampling respectively.

解析:

Explanation:

- 有放回抽样：样本空间 16 个样本点，事件A含 4 个，$P(A)=\frac{1}{4}$；With-replacement sampling: The sample space has 16 sample points, event A contains 4 of them, so $P(A)=\frac{1}{4}$;
- 不放回抽样：样本空间 12 个样本点，事件A含 2 个，$P(A)=\frac{1}{6}$；Without-replacement sampling: The sample space has 12 sample points, event A contains 2 of them, so $P(A)=\frac{1}{6}$;
- 分层抽样：样本空间 4 个样本点，事件A为空集，$P(A)=0$。Stratified sampling: The sample space has 4 sample points, event A is an empty set, so $P(A)=0$.